Unit PoE-P4

SKU:U213



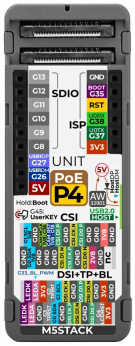


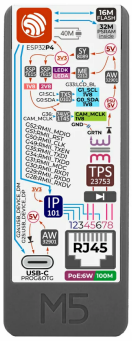




















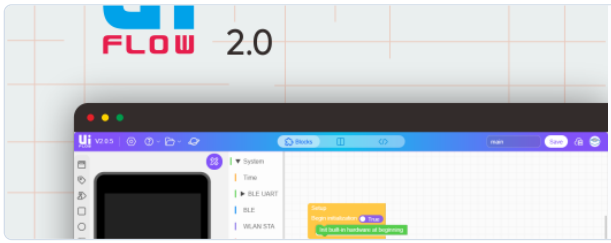
Description

Unit PoE-P4 is a high-performance PoE Ethernet embedded controller powered by the **ESP32-P4** SoC (RISC-V 32-bit dual-core @ 360MHz + LP single-core @ 40MHz). It integrates **16MB Flash** and **32MB PSRAM**, and features the built-in **IP101GRI** Ethernet PHY supporting **10/100 Mbps** communication. The device also supports PoE power delivery with a maximum output of 6W, and includes an onboard DC-DC converter to supply system power, greatly simplifying power wiring in practical applications.

It is equipped with a 24P FPC display expansion interface (MIPI DSI + TP + BL) and a 24P FPC camera interface (MIPI CSI), supporting high-resolution display output and image capture. For peripheral expansion, it provides one USB Type-C Host (USB 2.0) port for connecting U disks, HID devices, and other peripherals, plus one USB Type-C (USB 2.0) port supporting OTG and firmware download for debugging. In addition, it features a Hat expansion interface, SDIO & ISP IO expansion interfaces, and a Grove interface, enabling flexible connection to a wide range of sensors and peripherals.

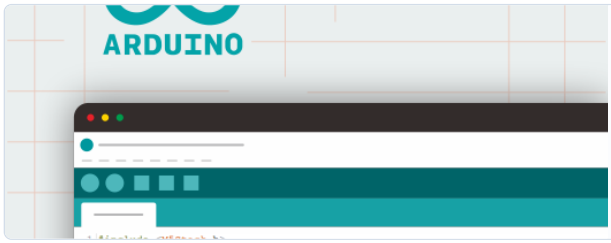
The device also integrates an RGB indicator LED, an infrared transmitter circuit, and a user button to support status indication, local interaction, and infrared signal transmission. It is ideal for applications such as industrial HMI, PoE smart terminals, vision acquisition nodes, smart home controllers, and edge computing devices.

Tutorial



UiFlow2

This tutorial introduces how to control Unit PoE-P4 through the UiFlow2 graphical programming platform.



Arduino IDE

This tutorial introduces how to program and control the Unit PoE-P4 using the Arduino IDE.

Features

- ESP32-P4 main controller:
  - 16MB Flash + 32MB PSRAM
- Peripheral interfaces:
  - 1x 24P FPC camera interface: MIPI CSI 2-Lane
  - 1x 24P FPC display expansion interface: MIPI DSI 2-Lane + TP + BL

    - 1x USB Type-C Host interface (USB 2.0)
    - 1x USB Type-C firmware download / OTG interface (USB 2.0)
- Ethernet:
    - Integrated IP101 10/100M Ethernet PHY chip
    - Supports PoE (active Ethernet) power supply, enabling data transmission and power delivery over a single Ethernet cable
- Human-machine interaction:
    - 1x RGB LED indicator
    - 1x user button
    - 1x infrared transmitter
- Expansion interfaces:
    - HY2.0-4P expansion interface
    - Hat2-Bus 2.54-16P
    - SDIO-Bus 2.54-9P
    - ISP-Bus 2.54-6P

Includes

- 1 x Unit PoE-P4
- 1 x HY2.0-4P Grove Cable (20cm)

Applications

- Industrial Human-Machine Interface (HMI)
- PoE smart terminals
- Vision acquisition nodes
- Smart home controllers
- Edge computing devices

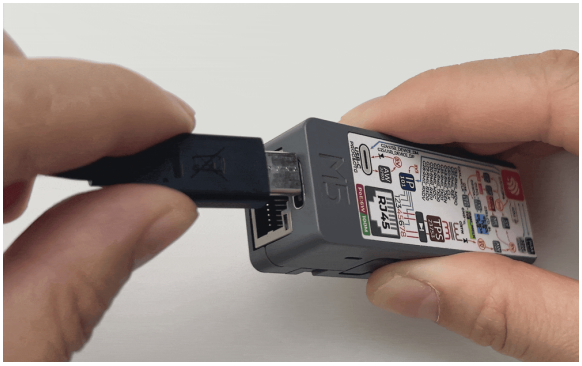
Specifications

Specification	Parameter
SoC	ESP32-P4NRW32 (RISC-V 32-bit high-performance dual-core processor @ 360 MHz + RISC-V 32-bit low-power single-core coprocessor @ 40 MHz)
Flash	16MB
PSRAM	32MB Octal
Power Input	DC 5V (USB Type-C) / PoE
Ethernet Controller	IP101GRI
Ethernet Speed	10/100M adaptive
Ethernet Interface	RJ45
PoE Standard	Compliant with IEEE 802.3at
PoE Power	Maximum power 6W
Display Interface	24P FPC (0.5mm pin pitch) MIPI DSI 2-Lane + TP + BL
Display Resolution	MIPI DSI display supports up to 1920×1080 (limited by lane count and frame rate)
Camera Interface	24P FPC (0.5mm pin pitch) MIPI CSI 2-Lane
USB Type-C	1x USB Type-C Host interface (USB 2.0) 1x USB Type-C firmware download / OTG interface (USB 2.0)
Power Consumption	Operating current: 5V @ 73.82mA Infrared transmission: 3.3V @ 23.57mA (instantaneous) RGB: 5V @ 75.32mA Full load (display, camera, IR, RGB, etc.): 277.10mA Deep sleep: 5V @ 19.85mA
RGB LED	1 x NH-B2020RGBA-HF
Infrared Transmitter	1 x MHS153IRCT
Operating Temperature	0 ~ 40°C
Product Size	64.0 x 24.0 x 20.2mm
Product Weight	28.2g
Package Size	65.0 x 33.0 x 21.2mm
Gross Weight	35.0g

Learn

Download Mode

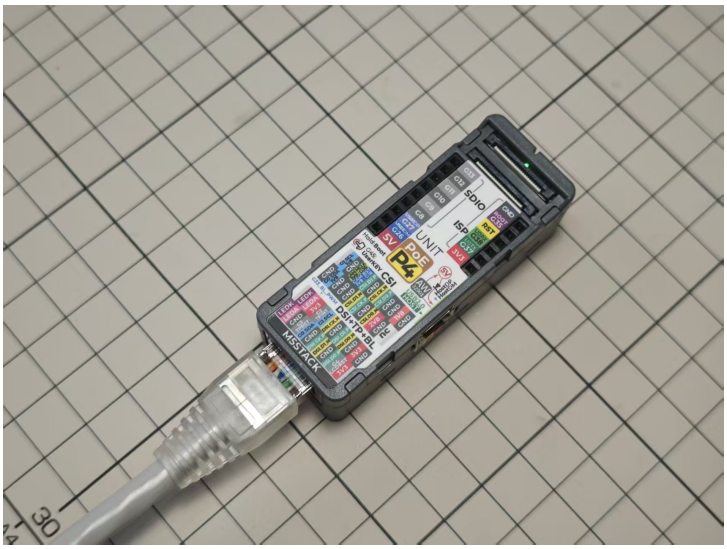
To flash firmware, connect the device to a computer via a USB Type-C data cable. Press and hold the side button for 3 seconds until the green LED lights up. The device will then enter download mode and wait for firmware flashing.



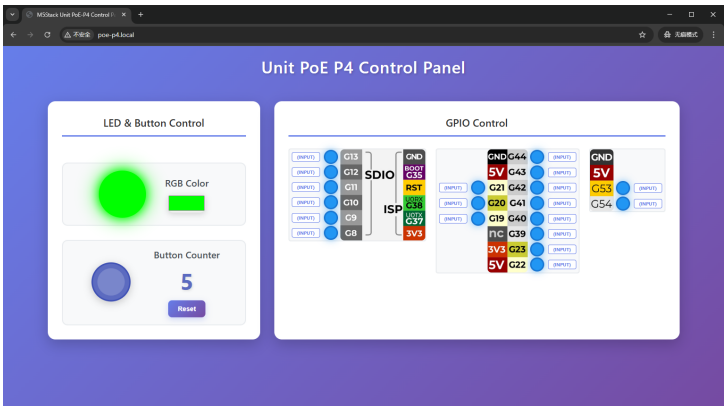
Access Device Control Page

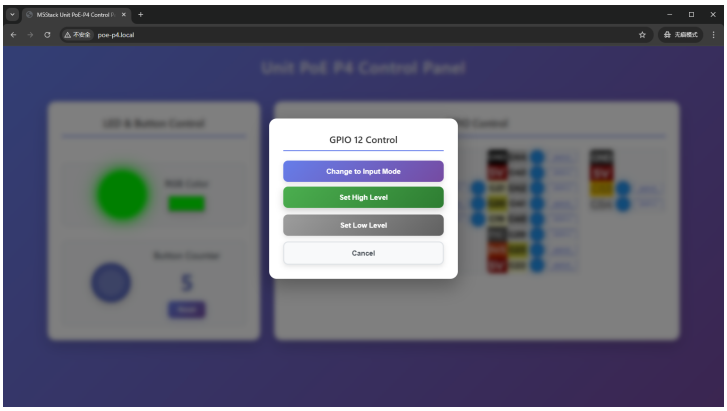
The factory default firmware of the device is preloaded with a WEB control panel, which supports RGB LED color control, button press count statistics, GPIO pin mode switching, and high/low level output. The device control page can be accessed by following these steps:

1. Power on the Unit PoE-P4 and connect it to the Ethernet network.
2. Open the serial monitor and note the device’ s current IP address.
3. Ensure your computer or mobile phone is on the same local area network (LAN) as the device.
4. Access the web control page using one of the following methods:



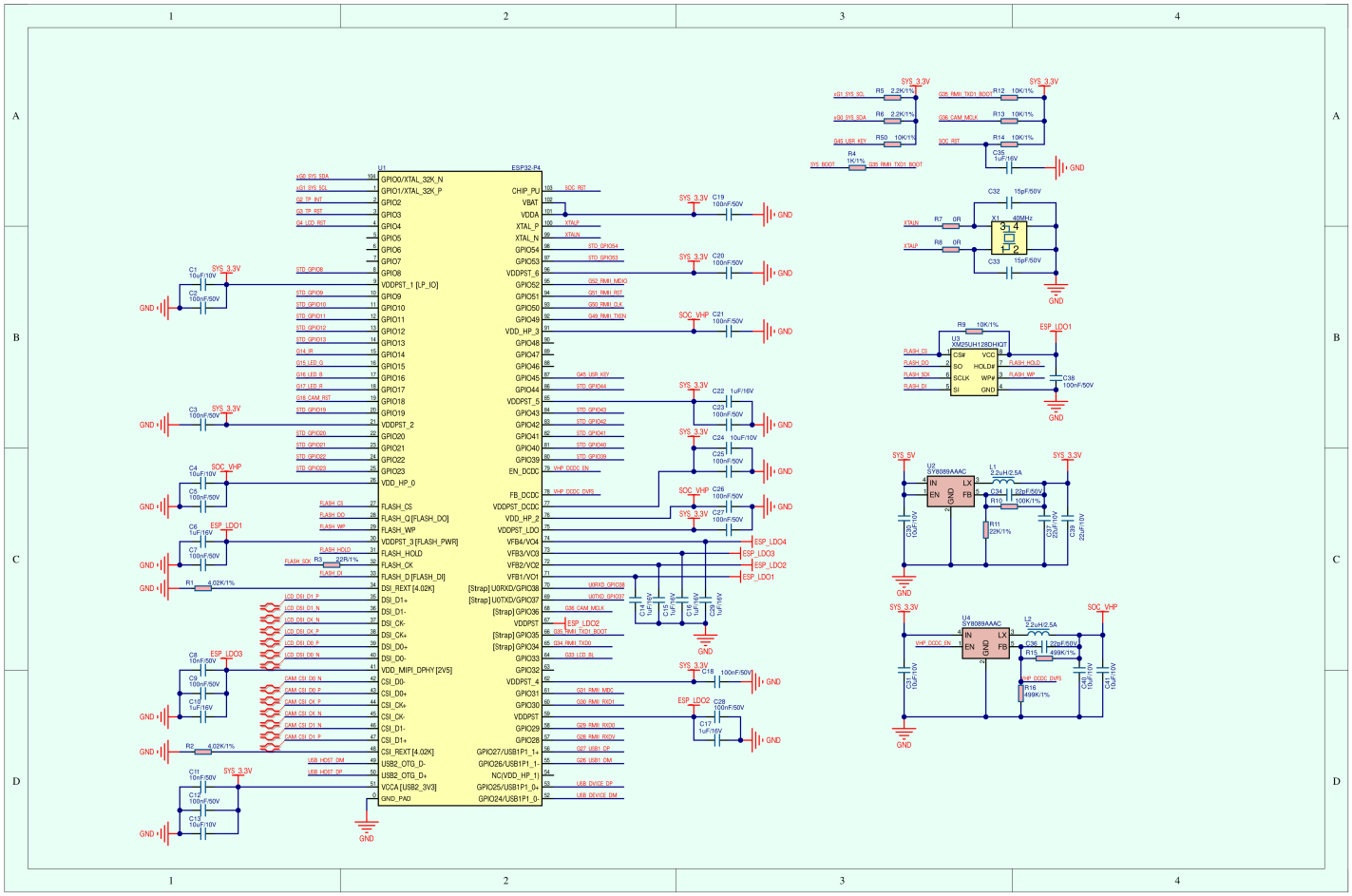
- Enter `poe-p4.local` in your browser
- Copy the IP address shown in the serial monitor and paste it into the browser address bar

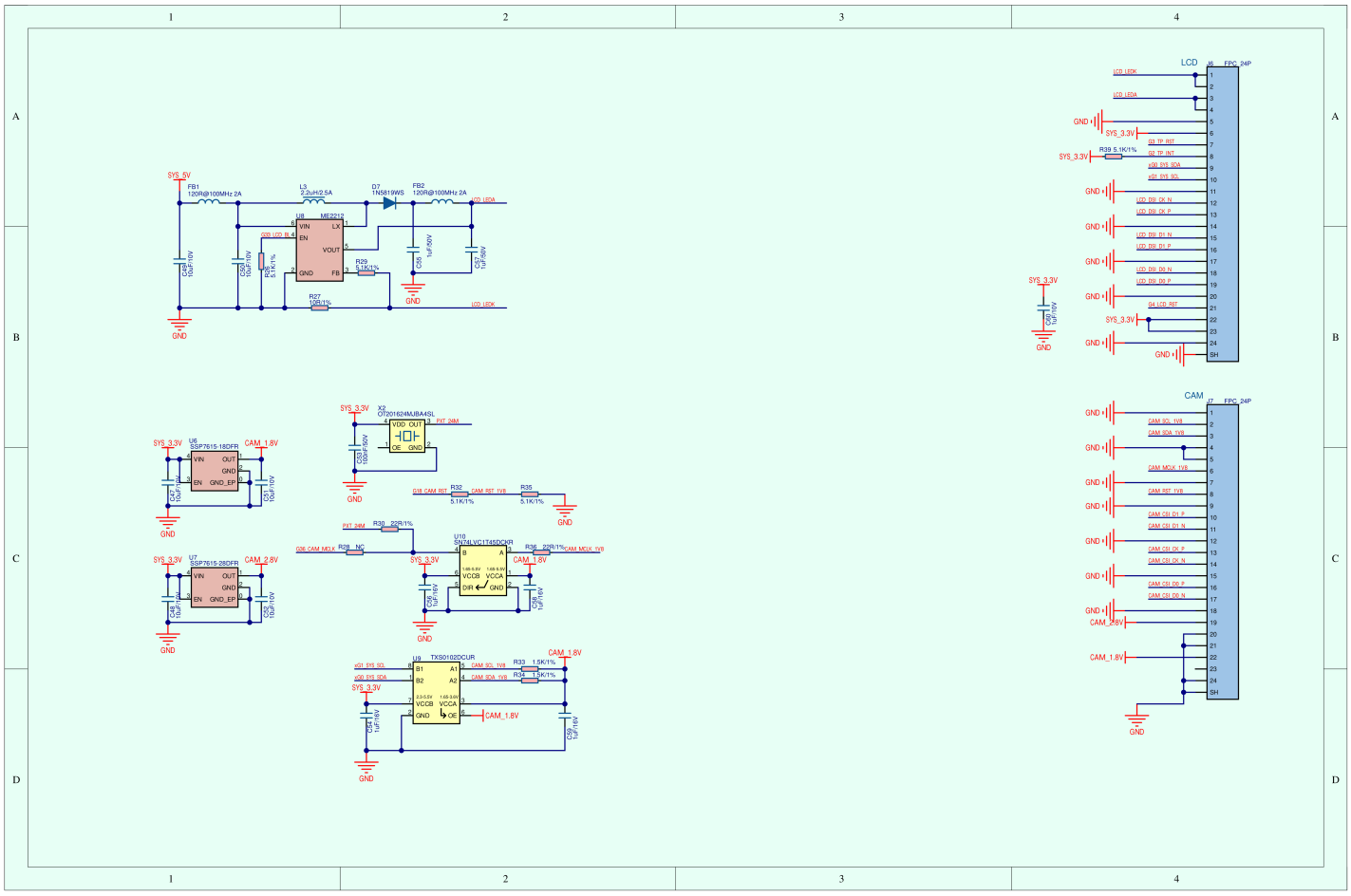


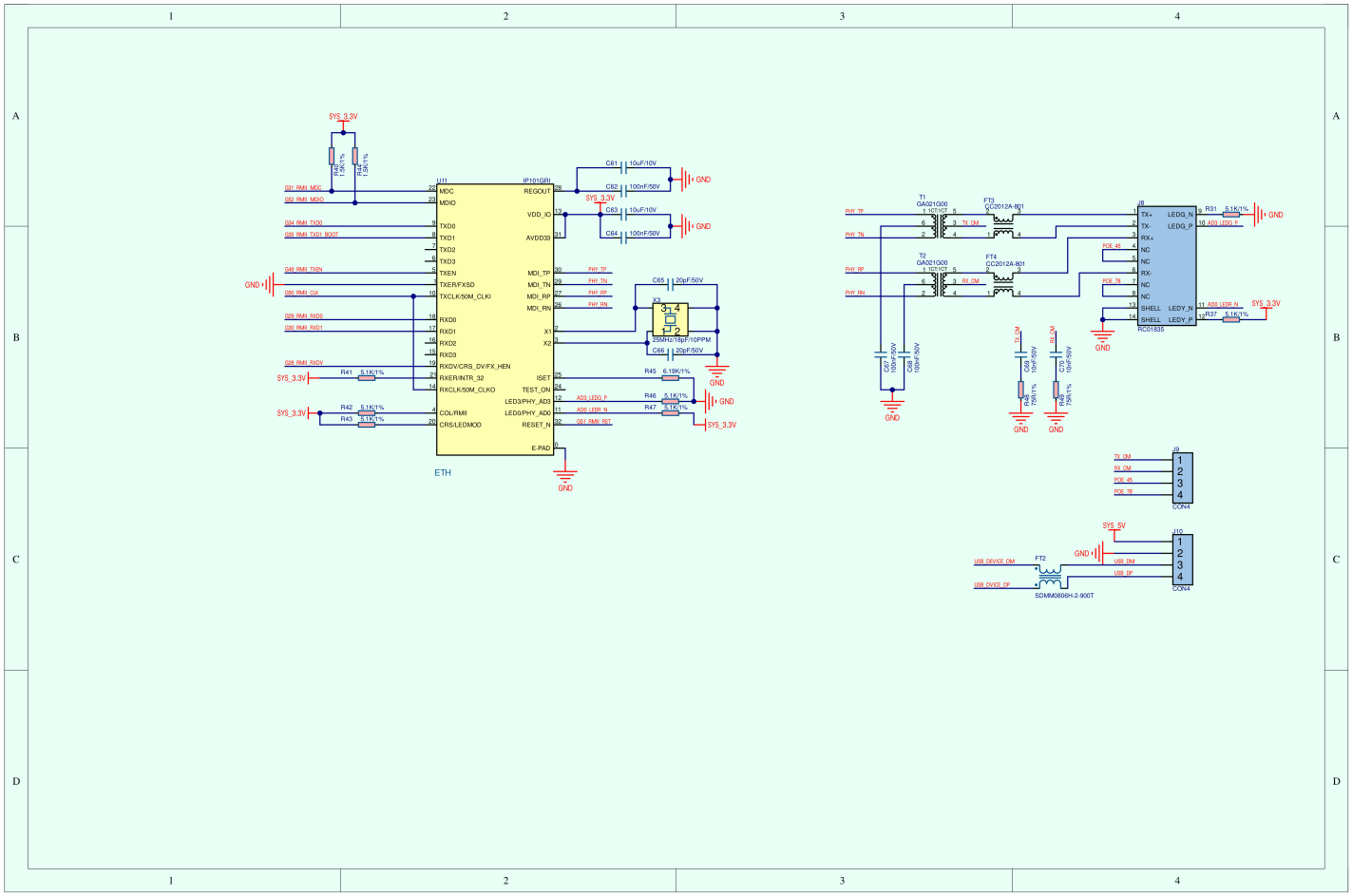


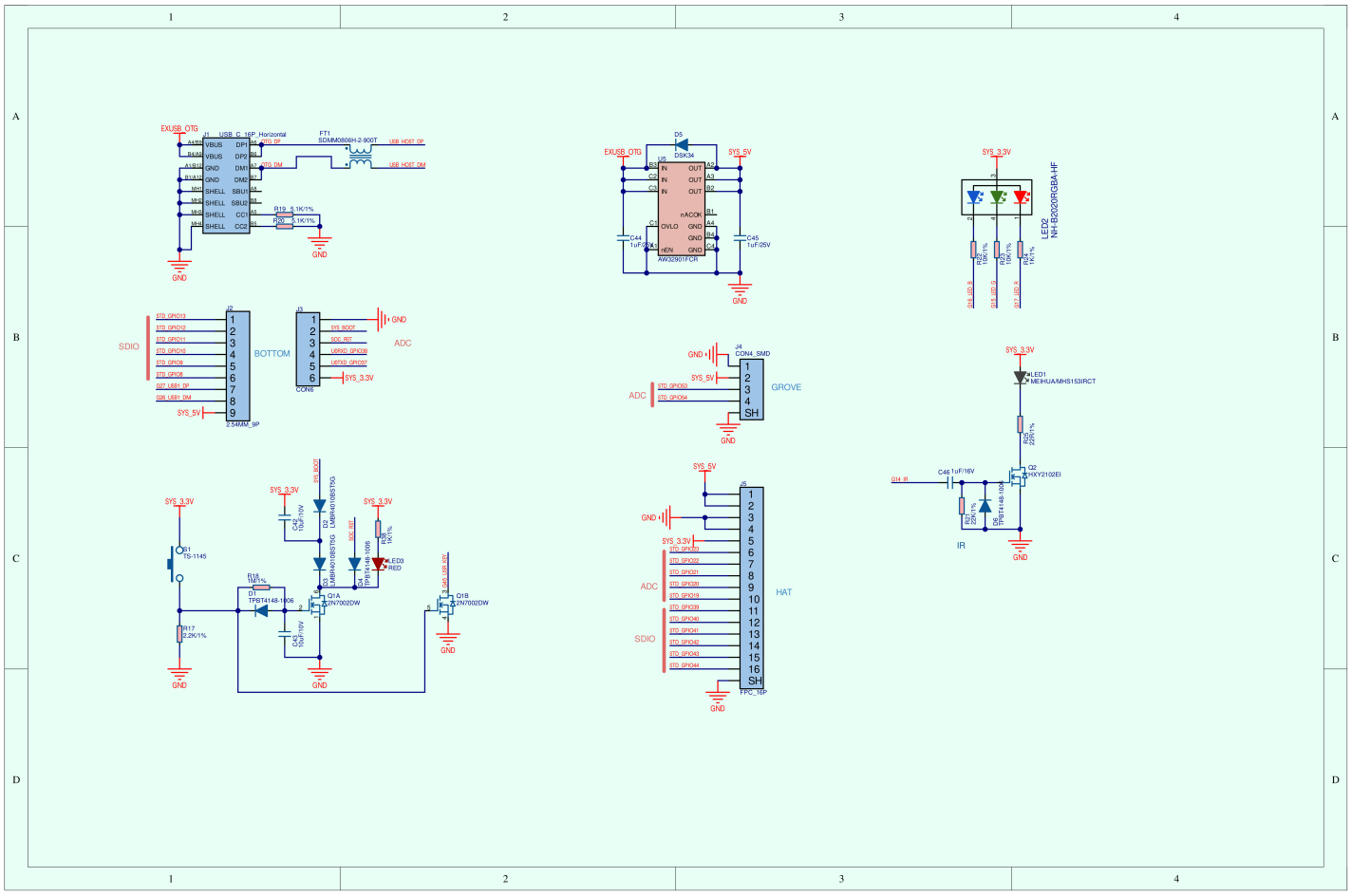
Schematics

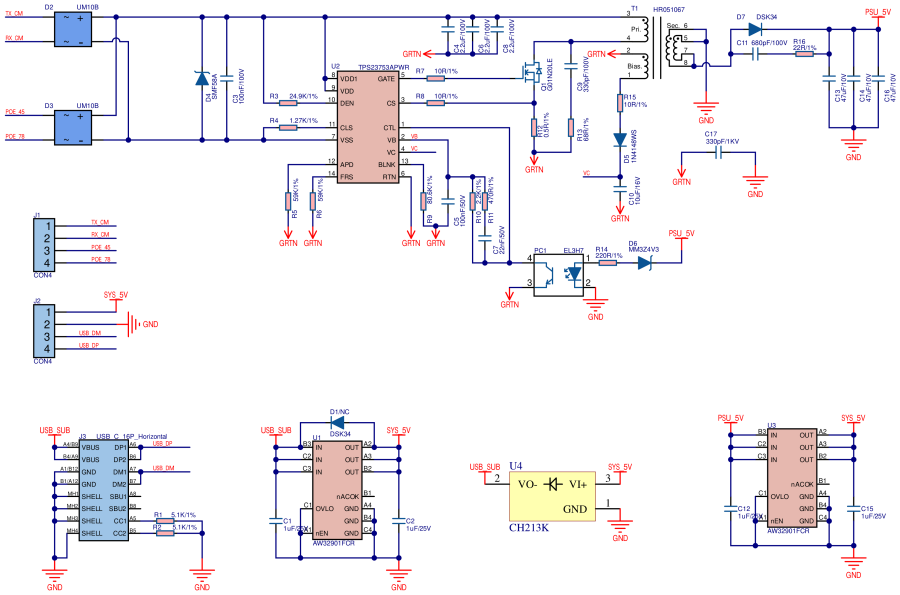
- Unit PoE-P4 Mainboard Schematics PDF
- Unit PoE-P4 Power Board Schematics PDF
- Unit PoE-P4 HAT Expansion Interface Schematics PDF

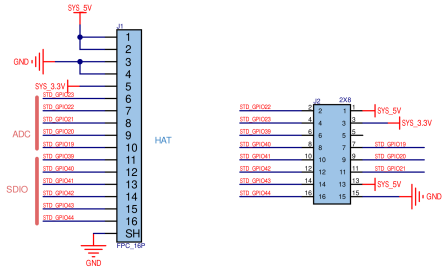

U1
ESP32-P4
GPIO0/XTAL_32K_N
GPIO1/XTAL_32K_P
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
VDDPST_1 [LP_IO]
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
VDDPST_2
GPIO20
GPIO21
GPIO22
GPIO23
VDD_HP_0
FLASH_CS
FLASH_Q[FLASH_DO]
FLASH_WP
VDDPST_3 [FLASH_PWR]
FLASH_HOLD
FLASH_CK
FLASH_D[FLASH_DI]
DSI_REXT [4.02K]
DSI_D1+
DSI_D1-
DSI_CK-
DSI_CK+
DSI_D0+
DSI_D0-
VDD_MIPI_DPHY [2V5]
CSI_D0-
CSI_D0+
CSI_CK+
CSI_CK-
CSI_D1-
CSI_D1+
CSI_REXT [4.02K]
USB2_OTG_D-
USB2_OTG_D+
VCCA [USB2_3V3]
GND_PAD
CHIP_PU
VBAT
VDDA
XTAL_P
XTAL_N
GPIO54
GPIO53
VDDPST_6
GPIO52
GPIO51
GPIO50
GPIO49
VDD_HP_3
GPIO48
GPIO47
GPIO46
GPIO45
GPIO44
VDDPST_5
GPIO43
GPIO42
GPIO41
GPIO40
GPIO39
EN_DCDC
FB_DCDC
VDDPST_DCDC
VDD_HP_2
VDDPST_LDO
VFB4/VO4
VFB3/VO3
VFB2/VO2
VFB1/VO1
[Strap] U0RXD/GPIO38
[Strap] U0TXD/GPIO37
[Strap] GPIO36
VDDPST_3
[Strap] GPIO35
[Strap] GPIO34
GPIO33
GPIO32
VDDPST_4
GPIO31
GPIO30
VDDPST
GPIO29
GPIO28
GPIO27/USB1P1_I
GPIO26/USB1P1_O
NC(VDD_HP_1)
GPIO25/USB1P1_Oi
GPIO24/USB1P1_Oi
SYS_3.3V
GND
SOC_VHP
ESP_LDO1
ESP_LDO2
ESP_LDO3
ESP_LDO4
U2
SY8089AAAC
U3
XMSJUH128DHIQT
U4
SY8089AAAC
X1
40MHz
FLASH_CS
FLASH_DO
FLASH_WP
FLASH_HOLD
FLASH_CK
FLASH_DI


LCD
J6
FPC_24P
CAM
J7
FPC_24P
U6
SSP7615-18DFR
U7
SSP7615-28DFR
U8
ME2212
U9
TXS0102DCUR
U10
SN74LVC1T45DCKR
X2
OT2016024MJBA4SL
SYS_5V
SYS_3.3V
CAM_1.8V
CAM_2.8V
GND


SYS_3.3V
ETH
GND
SYS_5V
USB_DEVICE_DM
USB_DEVICE_DP
USB_DM
USB_DP
FT2
SDMM0805H-2-900T
CON4


EXUSB_OTG
USB_C_16P_Horizontal
SDMM0806H-2-900T
SDIO
BOTTOM
ADC
CON6
2.54MM_9P
DSK34
AW3290TFCR
GROVE
CON4_SMD
HAT
FPC_16P
LED2
NHHB2020RGBA-HF
LED1
MEIHUA/MHS153IRCT
IR
TPBT414B-1006
2N7002DW
TS-1145


TX_DM
RX_DM
POE_45
POE_78
D2
UM10B
D3
UM10B
D4
SMF5V0A
C3
10uF/100V
U2
TPS23753APWR
VDD1
VDD
DEN
GATE
CS
CLS
VSS
APD
FRS
CTL
VB
VC
BLNK
RTN
R3 24.9K/1%
R4 1.27K/1%
R7 10R/1%
R8 10R/1%
R5 50K/1%
R6 50K/1%
R9 80.6K/1%
R10 4.99K/1%
R11 4.99K/1%
C4 0.1uF/100V
C5 0.1uF/100V
C6 2.2uF/100V
C7 0.1uF/100V
C8 330pF/100V
Q1 GRTN
R12 0.15R/1%
R13 0.68R/1%
C9 100pF/50V
C10 0.22uF/50V
GRTN
T1
HR051067
Pri.
Sec. 6
Bias
D7 DSK34
C11 680pF/100V
R16 22R/1%
PSU_5V
C13 47uF/16V
C14 47uF/16V
C16 47uF/16V
GND
R15 10R/1%
D5 1N4148WS
C12 10uF/16V
C17 330pF/1KV
VC
J1
CON4
J2
CON4
SYS_5V
USB_DM
USB_DP
PC1 EL3H7
R14 220R/1%
D6 MM3Z4V3
J3 USB_C_16P_Horizontal
USB_SUB
USB_DP
USB_DM
R1 5.1K/1%
R2 5.1K/1%
D1/NC
DSK34
U1
AW32901TFCR
C1 1uF/25V
C2 1uF/25V
U4
CH213K
VO-
VI+
U3
AW32901TFCR
C12 1uF/25V
C15 1uF/25V


SYS_5V
GND
SYS_3.3V
ADC
SDIO
HAT
J1
TPC_16P
SH
ZX8



PinMap

Ethernet

ESP32-P4	G31	G52	G51
IP101	SMI_MDC	SMI_MDIO	PHY_RST

ESP32-P4	G28	G29	G30	G34	G35	G49	G50
IP101	RMII_RXDV	RMII_RXD0	RMII_RXD1	RMII_TXD0	RMII_TXD1	RMII_TXEN	RMII_CLK

LCD

ESP32-P4	G4	G33	DSI_CLK_N	DSI_CLK_P	DSI_D1_N	DSI_D1_P	DSI_D0_N	DSI_D0_P
FPC - MIPI DSI	LCD_RST	LCD_BL	LCD_DSI_CK_N	LCD_DSI_CK_P	LCD_DSI_D1_N	LCD_DSI_D1_P	LCD_DSI_D0_N	LCD_DSI_D0_P

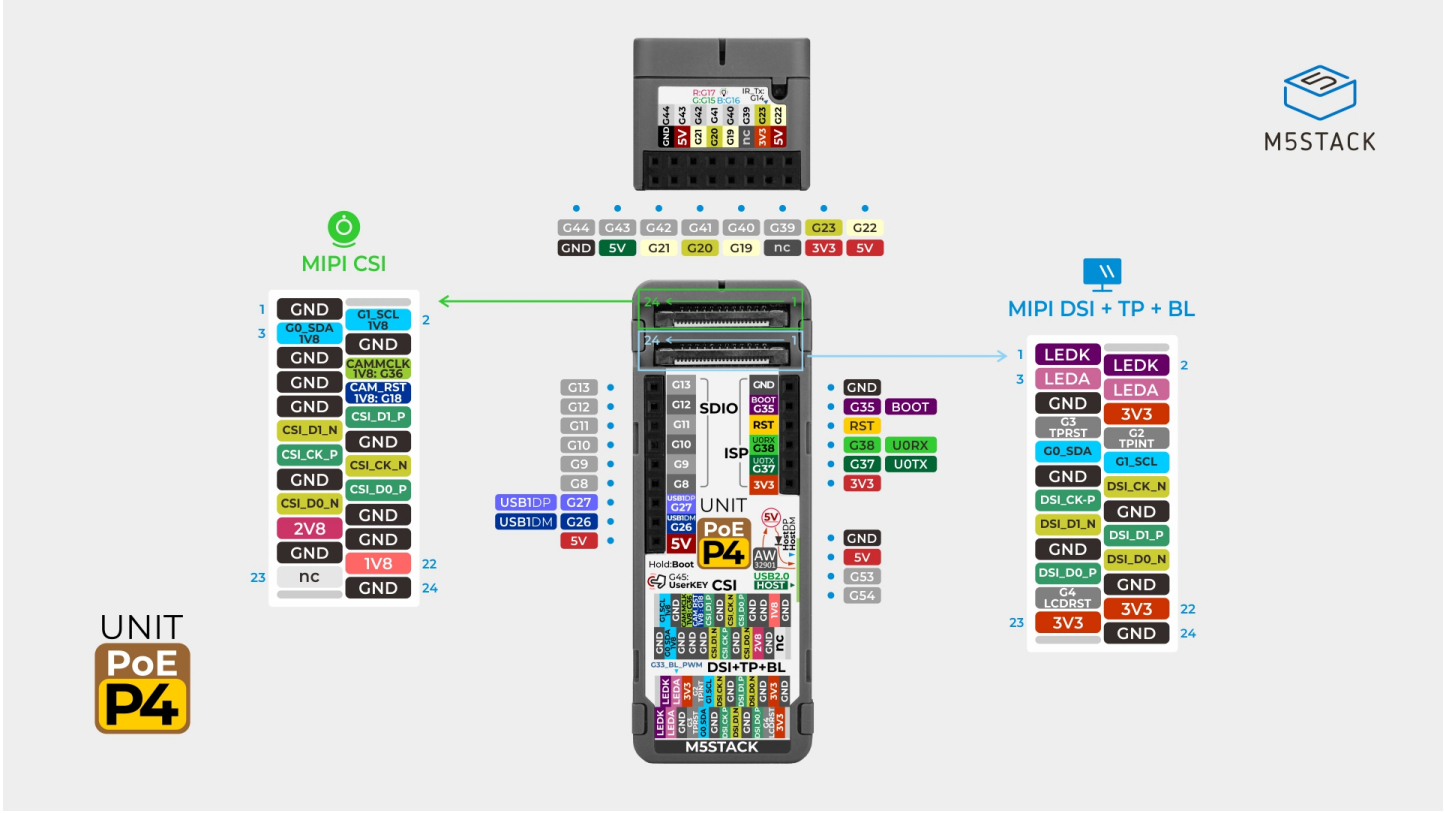
TP

ESP32-P4	G0	G1	G3	G2
FPC - TP	SDA	SCL	TP_RST	TP_INT

Camera

ESP32-P4	CSI_CK_N	CSI_CK_P	CSI_D1_N	CSI_D1_P	CSI_D0_N	CSI_D0_P
MIPI CSI FPC	CAM_CSI_CK_N	CAM_CSI_CK_P	CAM_CSI_D1_N	CAM_CSI_D1_P	CAM_CSI_D0_N	CAM_CSI_D0_P

ESP32-P4	G0	G1	G18	G36
MIPI CSI FPC	SDA_1V8	SCL_1V8	CAM_RST	CAM_MCLK



USB Type-C

ESP32-P4	USB2_OTG_DP	USB2_OTG_DN
USB 2.0 HOST	HOST_DP	HOST_DN

ESP32-P4	G25	G24
USB 2.0 Download / OTG	USB0_DP	USB0_DN

IR & RGB LED & Button

ESP32-P4	G14	G15	G16	G17	G45
IR	IR_TX				
RGB LED		LED_GREEN	LED_BLUE	LED_RED	
Button					KEY_USER

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	G53	G54

SDIO & ISP

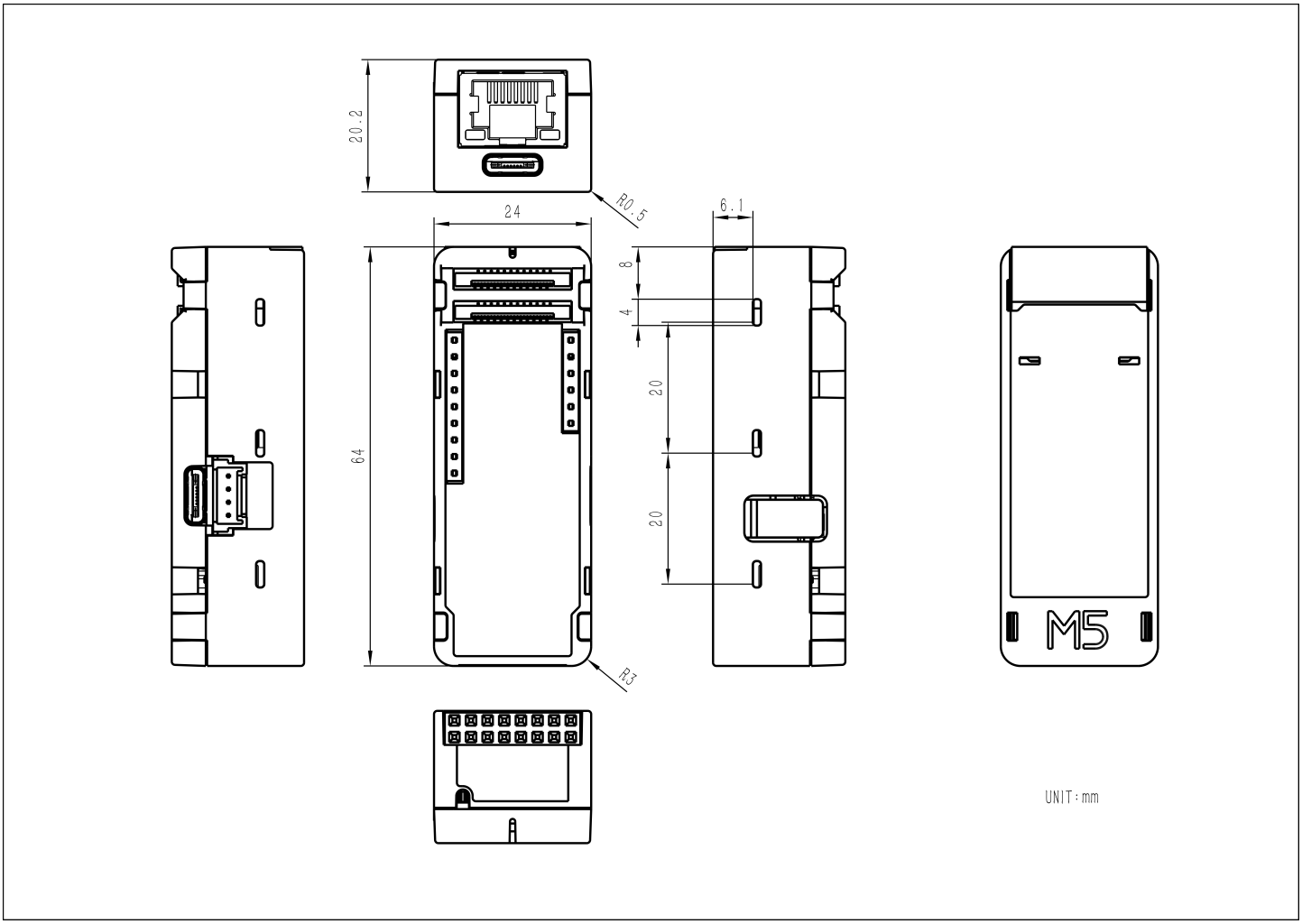
FUNC	PIN	LEFT	RIGHT	PIN	FUNC
	G13	1	1	GND	
	G12	2	2	G35	BOOT
	G11	3	3	RST	
	G10	4	4	G38	UART0_RX
	G9	5	5	G37	UART0_TX
	G8	6	6	3V3	
USB1_DP	G27	7			
USB1_DM	G26	8			
	5V	9			

Hat2-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	G44
5V	3	4	G43
G21	5	6	G42
G20	7	8	G41
G19	9	10	G40
NC	11	12	G39
3V3	13	14	G23
5V	15	16	G22

Model Size

- Unit PoE-P4 Model Size PDF



Structure

- Unit PoE-P4 Structure Files

Datasheets

- ESP32-P4
- IP101GRI
- TPS23753A

Softwares

Arduino

- Unit PoE-P4 Arduino Quick Start

UiFlow2

- Unit PoE-P4 UiFlow2 Quick Start

ESP-IDF

- Unit PoE-P4 Factory Firmware Source Code
- Unit PoE-P4 Factory Firmware Build Guide

Other

- Unit PoE-P4 Factory Reset Firmware Tutorial

Video